1. Identity

Document	STAR-PKG-CONJUNCTION-SCREEN-SGP4-EN · v2.2 · 2026-11-02
Package	conjunction-screen-sgp4
Version	0.2.1
Capability	star.conjunction.screen
Maturity	Beta (VERIFIED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/conjunction_screen_sgp4

2. Licence and obligations

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3. What it needs at run time

NEEDS	LICENCE	SOURCE
numpy	BSD-3-Clause	https://numpy.org/
scipy	BSD-3-Clause	https://scipy.org/
sgp4	MIT	https://pypi.org/project/sgp4/

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())"
conjunction_screen_sgp4-0.2.1-py3-none-any.whl
```

FILE	SHA-256
conjunction_screen_sgp4-0.2.1-py3-none-any.whl	b1612b180d8a85a7395578f7623432b8b33583679f350f33fc49990ee61fce08
conjunction_screen_sgp4-0.2.1-source.zip	ec18fcb5a014934f24ddda378105beb03718fb2ad777fd6863ded03ce05f0f20

5. What it does, and its limits

Screening of close approaches between objects of TLE catalogues (who to look at), NOT a collision-probability tool.

Probability of collision, covariance, manoeuvre advice, accuracy below the ~1 km TLE error. Advisory screening only.

- `screen(tle_paths, start, hours, step, coarse_km, final_km, cross_only, refine)` returns encounters (TCA in UTC, miss distance km, age of the older TLE) below `final_km`, excluding pairs that stay close (formations).
- TLEs are propagated with SGP4 only (mean elements); propagation errors are reported, never replaced.
- `refine_tca(state_at, step)`: closed-form TCA from relative motion, shared by screening and tests.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_conjunction_screen_sgp4_0_2_1,
  author  = {Cavazzini, Andrea},
```

```
  title   = {{conjunction-screen-sgp4}: Conjunction screening of TLE catalogues with SGP4, KD-
tree coarse pass and closed-form TCA refinement},
  version = {0.2.1},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/conjunction_screen_sgp4/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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Related: space activities and export control · acceptable use · evidence and limits · licence texts